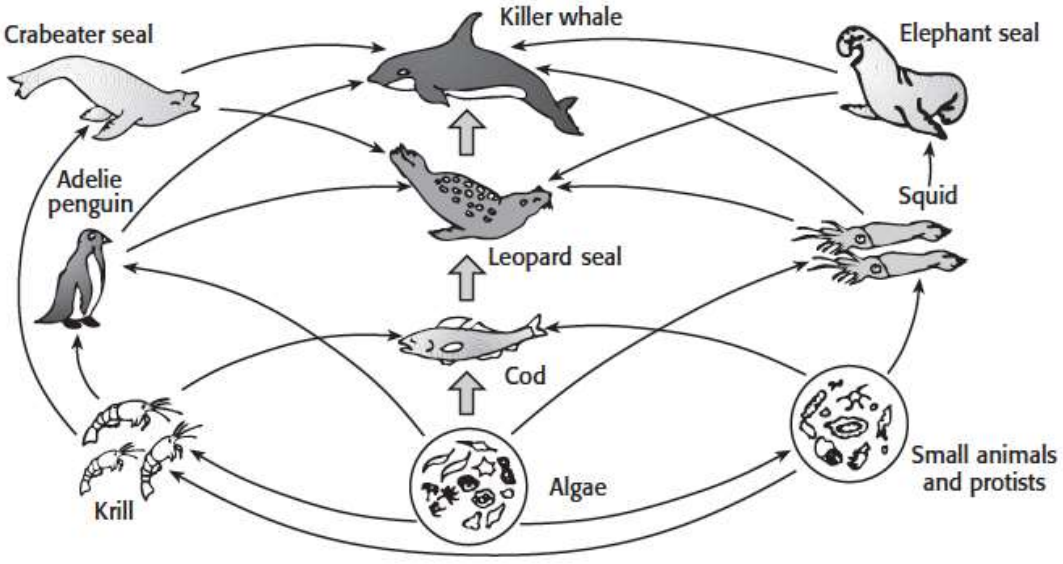
Name _____ Class _____ Date _____

Skills Worksheet

# Food Chains and Food Webs

## INTERPRETING GRAPHICS

**Use the figure below, which shows the food web of an aquatic ecosystem, to complete items 1–7.**




**In the food web above, there are eight food chains that include krill. In the space provided, identify all of the organisms in the order in which they occur in four of these eight food chains.**

**1.** Chain 1 _____

_____

**2.** Chain 2 _____

_____

**3.** Chain 3 _____

_____

**4.** Chain 4 _____

_____

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**Read each question about the food web on the previous page, and write your answer in the space provided.**

**5.** What organisms do cod eat?

_____

_____

**6.** List all the organisms that eat squid.

_____

_____

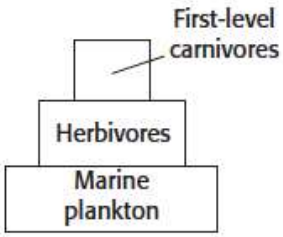
**7.** How many producers are in the food web? Name them.

_____

_____

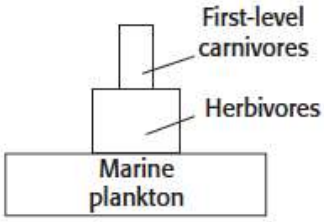
**Use the figures below, which show trophic levels in an ecosystem, to complete items 8–11.**

**Done by Counting Organisms at Each Level**



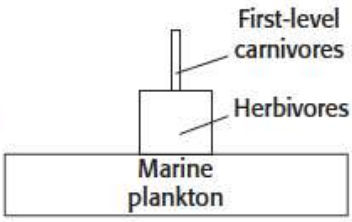

**A**

**Done by Weighing Organisms at Each Level**




**B**

**Done by Measuring the Calories Stored at Each Level**




**C**

**Study the three pyramids above. In the space provided, identify which pyramid is the most accurate indicator of each item below by writing the correct letter (*A–C*) in the space provided.**

_____ **8.** number of individual organisms

_____ **9.** measurement of productivity

_____**10.** measurement of biomass

**11.** Which pyramid is the most accurate indicator of the amount of energy available at each trophic level? Explain.

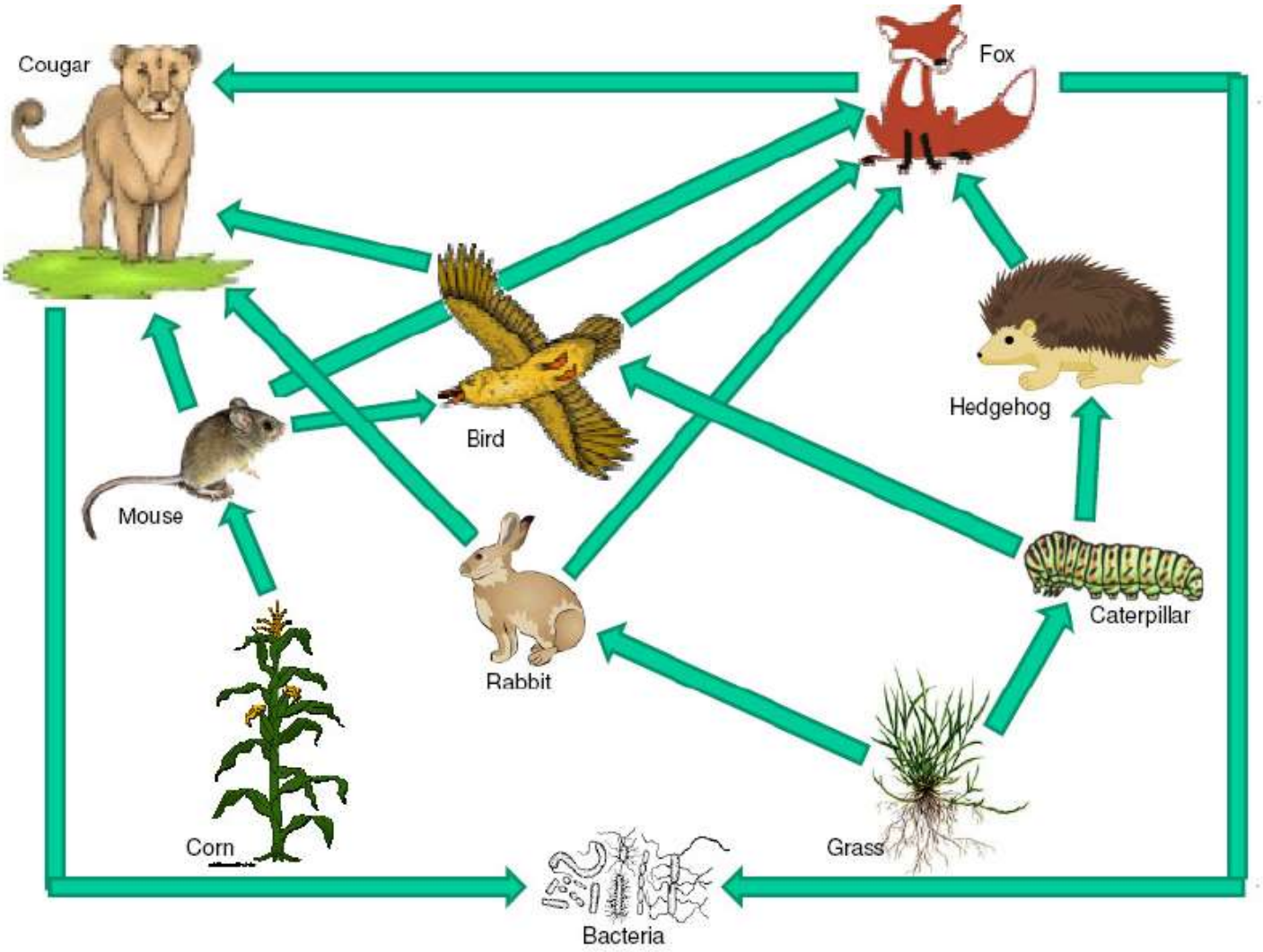
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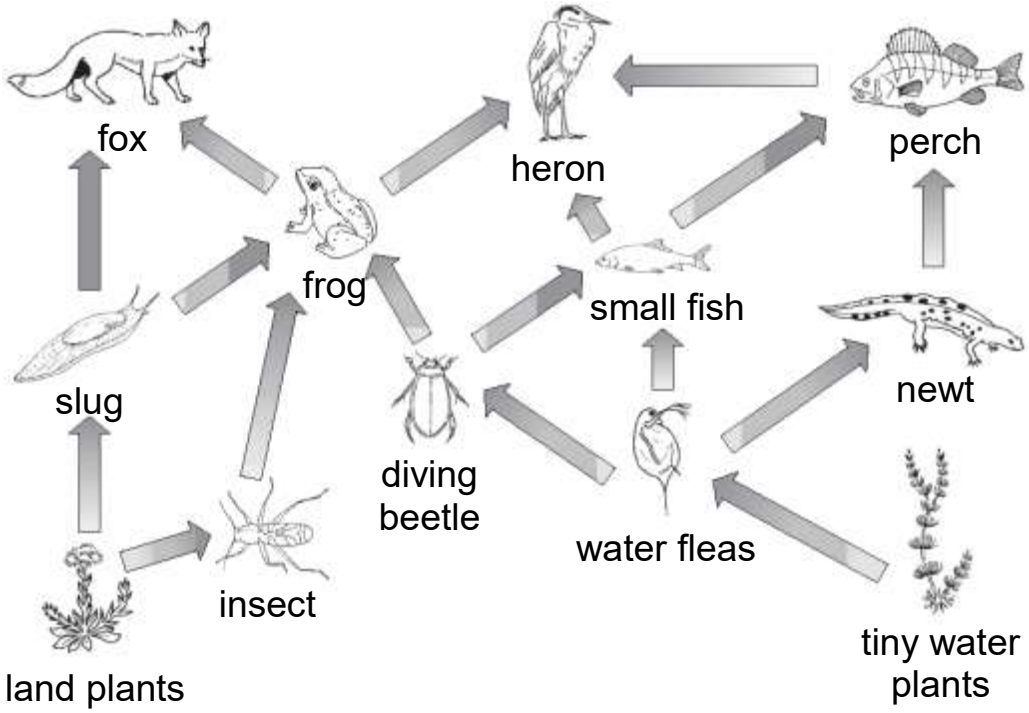
**Look at this food web. Then answer questions 12-18.**




From ShellysScienceSpot.com

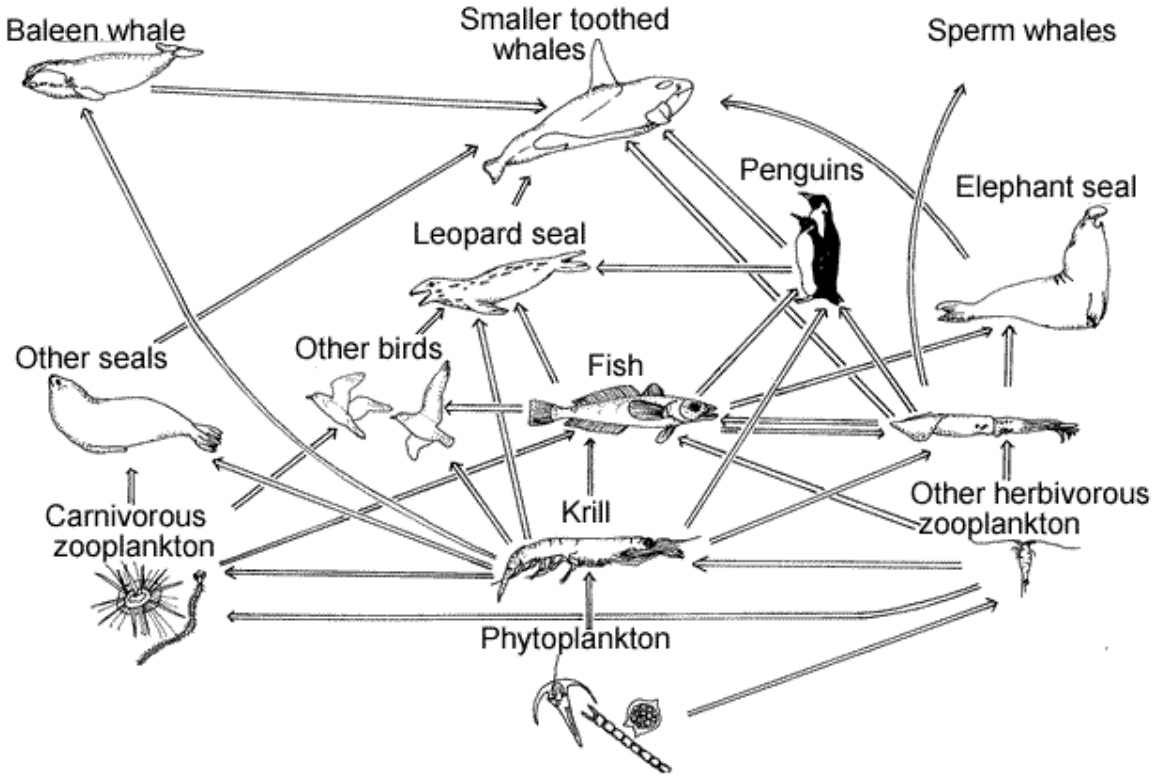
12) List the producers shown in the food web.
13) List the consumers shown in the food web.
14) List the decomposers shown in the food web.
15) List one food chain that is part of this food web (use arrows to show the transfer of energy).
16) Which animals in the food web are the primary (first level) consumers?
17) Which animals in the food web are the tertiary (third level) consumers?
18) Imagine that a landscaper was frustrated by caterpillars eating the grass of the lawns that he cared for The landscaper decided to use pesticide (a chemical that destroys pests), and successfully killed all of the caterpillars in the ecosystem. What organisms would be affected by the removal of caterpillars in the food web? Explain your answer.

**Look at this food web. Then use it to answer questions 19-25.**




19) Name two producers in the food web.
20) Name three consumers in the food web.
21) Name the animals that the small fish eats.
22) Name the animals that eat the small fish.
23) Explain what could happen to the community if all the frogs suddenly died.
24) How are food webs different to food chains? Explain why food webs are more useful.

**Look at this food web. Then use it to answer questions 27-33.**




25) Which organisms are the producers?
26) Which organisms are the primary consumers?
27) Which organisms are the secondary consumers?
28) Which organisms are the tertiary consumers?
29) List all the herbivores.
30) List 4 unique food chains from this food web:
    a)
    b)
    c)
    d)
31) Explain in detail what would happen to the food chain if a toxin released by humans killed all the phytoplankton.

# KEY

## Food Chains and Food Webs

### INTERPRETING GRAPHICS

**1–4.** Answers may include any of the following food chains:

Chain 1—algae, krill, Adelie penguin, leopard seal, killer whale
Chain 2—algae, krill, cod, leopard seal, killer whale
Chain 3—algae, small animals and protists, krill, crabeater seal, killer whale
Chain 4—algae, small animals and protists, krill, Adelie penguin, killer whale
Chain 5—algae, small animals and protists, krill, crabeater seal, leopard seal, killer whale
Chain 6—algae, krill, crabeater seal, leopard seal, killer whale
Chain 7—algae, krill, crabeater seal, killer whale
Chain 8—algae, krill, Adelie penguin, killer whale

**5.** krill, algae, small animals, and protists
**6.** leopard seal, killer whale, and elephant seal
**7.** one; algae
**8.** A
**9.** C
**10.** B
**11.** B; each higher level on the pyramid contains only 10 percent of the biomass found in the trophic level below it. Ecologists measure biomass to determine the amount of energy present in a trophic level.

12) Grass, corn
13) Mouse, rabbit, caterpillar, bird, hedgehog, fox, lion
14) Bacteria
15) There are many possible answers to this question. Ask your teacher if you are unable to answer it.
16) Mouse, rabbit, caterpillar
17) Fox, lion
18) Assume a closed system (no new animals may arrive or leave)
    a. Grass would increase (fewer consumers), rabbits would increase, lions may or may not increase (depends on importance of bird population in diet)
    b. Birds might decrease (depends on importance of caterpillars in diet), mice would increase (fewer birds to eat them), lions/foxes may decrease (depends on importance of caterpillars in diet.
    c. Hedgehogs would go extinct (no food), foxes may decrease (depends on importance of caterpillars in diet)
19) Tiny water plants, land plants
20) All answers are correct except for water plants or land plants
21) Water fleas and diving beetles
22) Heron and perch
23) Many possible answers:
    a. Beetle, slug, & insect populations would increase (fewer predators to eat them)
        i. Therefore, land plant and water flea population would decrease (more animals eating them)
        ii. Therefore, tiny water plant population would increase (no water fleas eating them)
    b. Herron and fox populations would decrease (fewer prey available)
        i. Therefore, small fish, perch, and slug populations would increase (fewer predators to eat them)
        ii. Therefore, land plants, driving beetle, water flea, and newt populations would decrease (more consumers eating them)
24) Food webs are more complex and therefore more accurately represent the relationships between living things.
25) Phytoplankton
26) Krill, herbivorous zooplankton
27) **Things that eat krill** (carnivorous zooplankton, other seals, baleen whale, other birds, leopard seal, fish, penguins, squid) and **things that eat herbivorous zooplankton** (fish, squid, krill, carnivorous zooplankton)
*(NOTE: Krill is BOTH a primary consumer and a secondary consumer because it eats phytoplankton, a producer, and zooplankton, a primary consumer)*

28) Many answers! Reasons listed below. **The total list is here: other birds, leopard seal, penguins, elephant seal, squid, sperm whales, penguins, fish, smaller toothed whales, carnivorous zooplankton, baleen whale, other seals**

a. Eaters of fish (other birds, leopard seal, penguins, elephant seal, squid)
b. Eaters of squid (elephant seal, sperm whales, penguins, fish, smaller toothed whales)
c. Eaters of krill (carnivorous zooplankton, other seals, baleen whale, other birds, leopard seal, fish, penguins)
d. Eaters of carnivorous zooplankton (other seals, other birds, fish)
e. Eaters of other seals (smaller toothed whales)
f. Eaters of birds (leopard seal)
g. Eaters of leopard seals (smaller toothed whales)
h. Eaters of penguins (leopard seals, smaller toothed whales)

*29)* Herbivorous zooplankton only.

*NOTE: Krill is not an herbivore because it also eats zooplankton which is an animal.*

30) There are many possible answers to this question. Ask your teacher if you are unable to answer it.

31) Phytoplankton would die out. All primary consumers would die out because they had no food, which would kill all secondary consumers, tertiary consumers, and quaternary consumers. Basically, everything would die.